



TRIGG: BUILD GUIDE

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WORKISH.BERLIN
CREATIVE COWORKING



1: COMPONENTS

Front Side

8 x 12mm Tactile Switches (buttons)

1 x 1M Resistor

(Brown Black Black Yellow Brown)

1 x 100k Resistor

(Brown Black Black Orange Brown)

2 x 1k Resistors

(Brown Black Black Brown Brown)

2 x 1uF Ceramic Disc Capacitor (105)

1 x Mini Slide Switch

2 x 3mm LEDs

1 x 1N5819 Diode

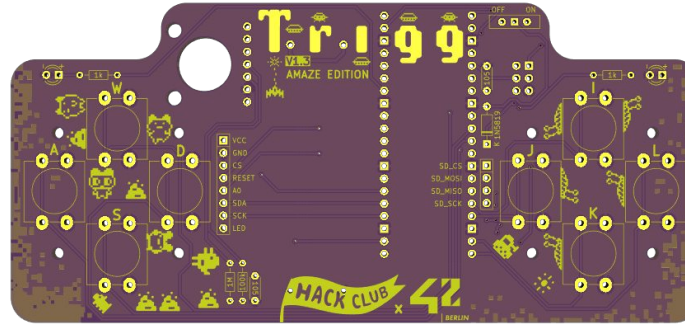
1 x 1.8" TFT Display

1 x 4-pin Header

1 x 4-pin Socket

1 x 8-pin Socket

Printed Circuit Board



Back Side

1 x Raspberry Pi Pico

2 x 20-pin Headers

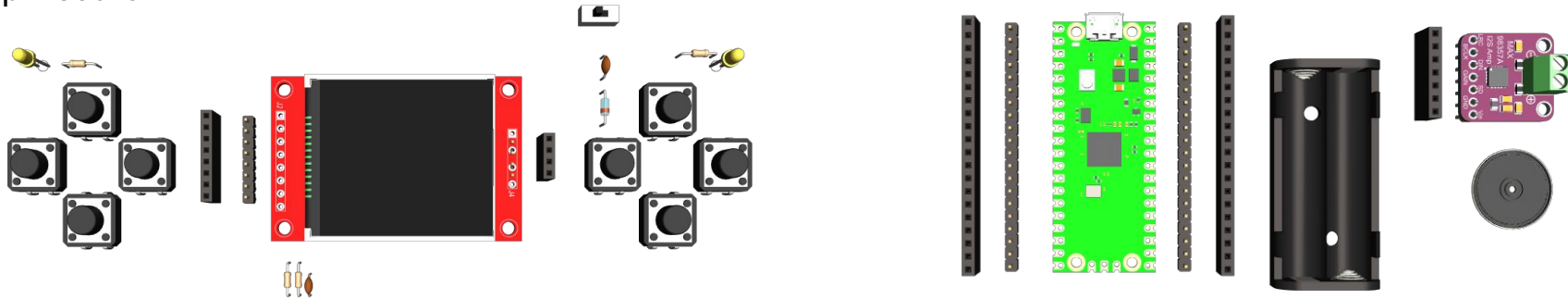
2 x 20-pin Sockets

1 x 2xAAA Battery Holder

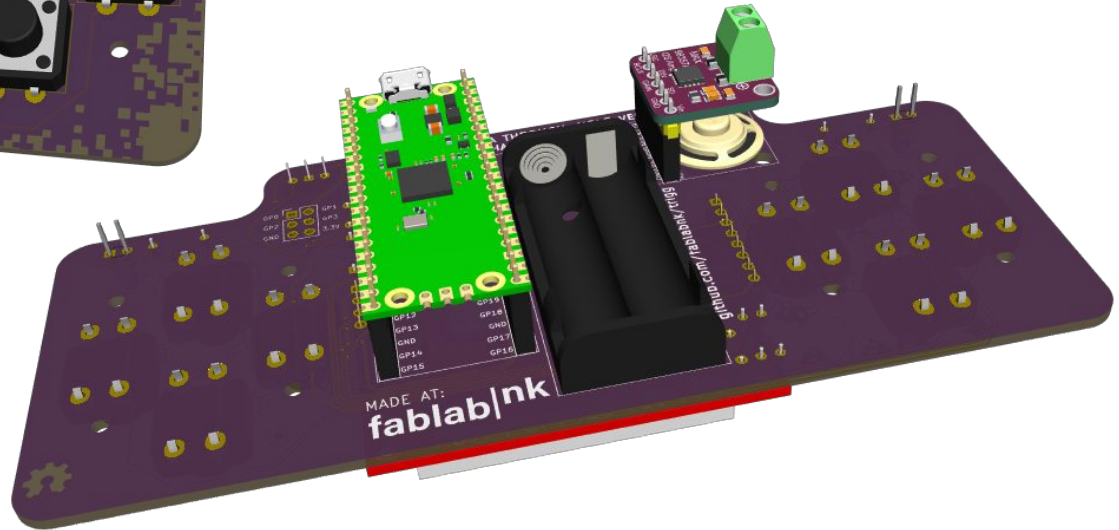
1 x Max98357A Amplifier Board

1 x 7-pin Socket

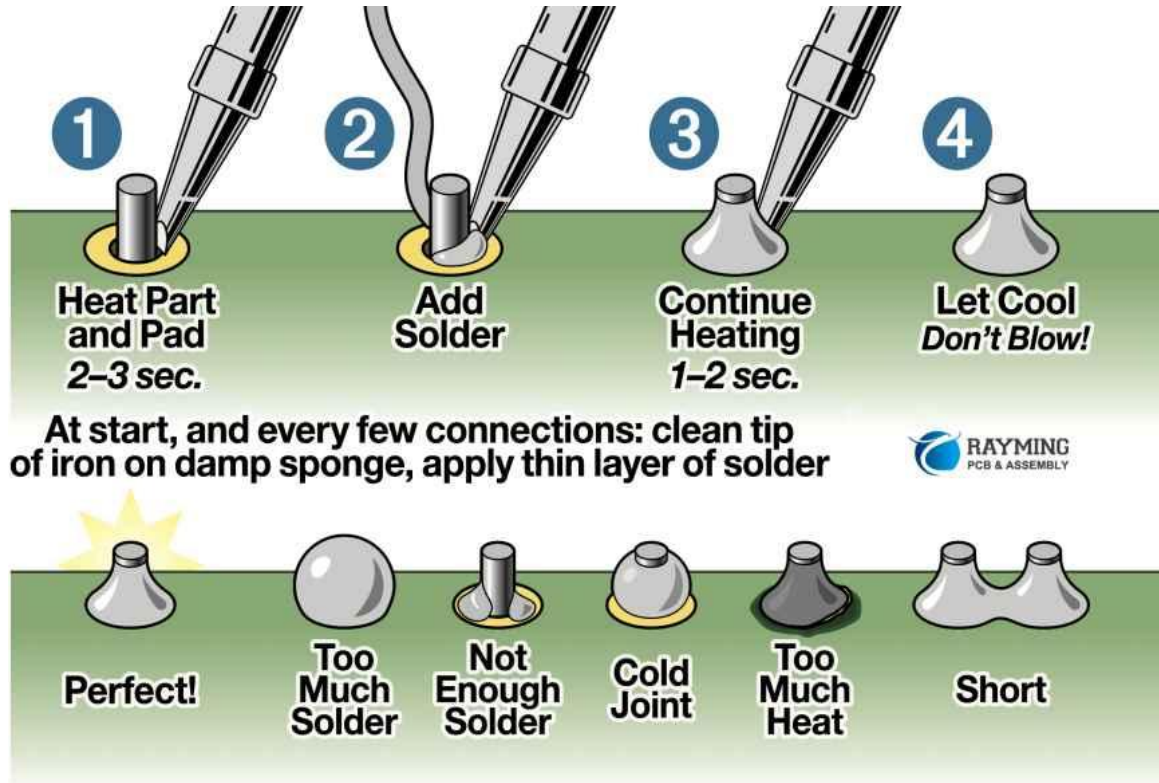
1 x 1W Speaker



2: FINISHED BUILD (HOW IT WILL LOOK AT THE END)

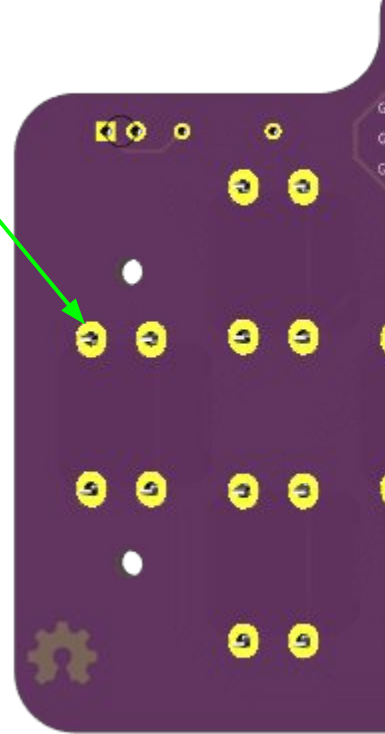
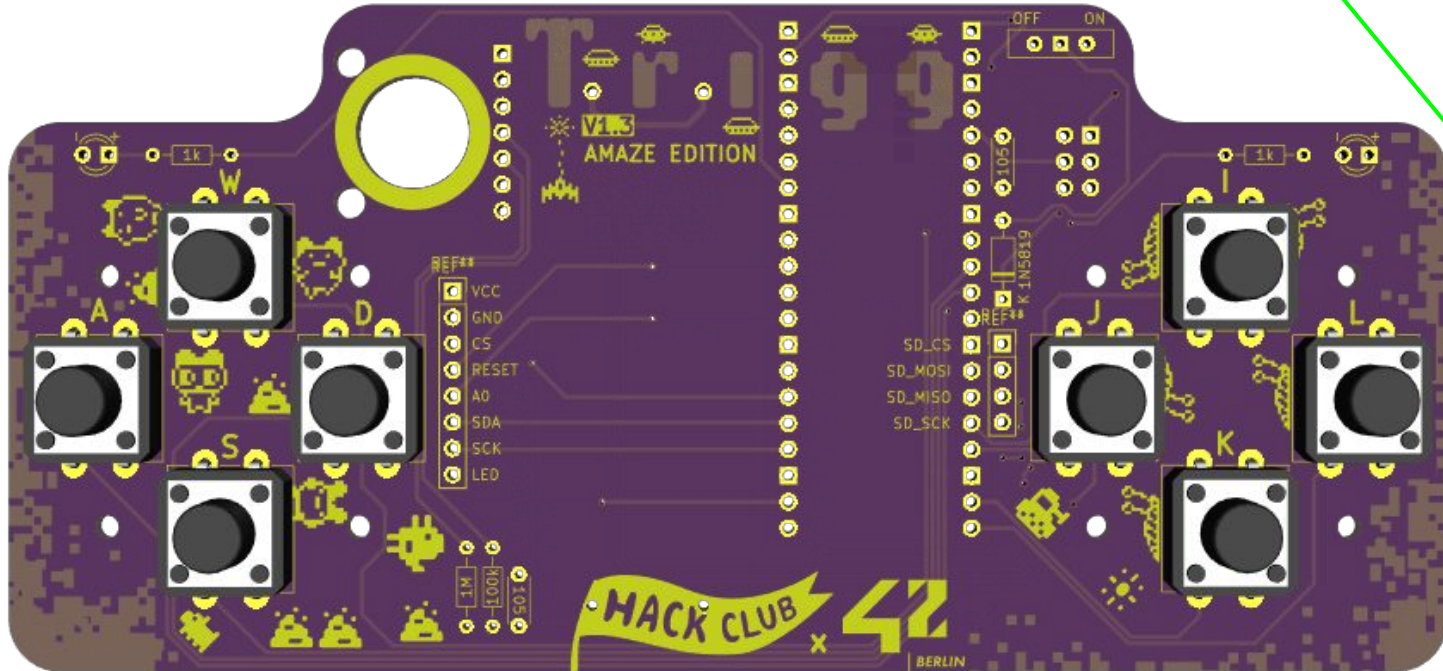


3: HOW TO SOLDER



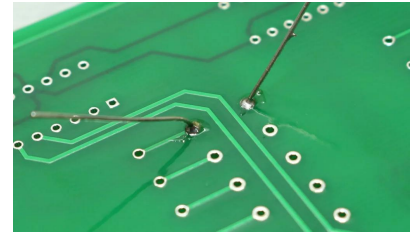
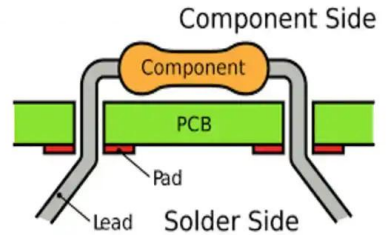
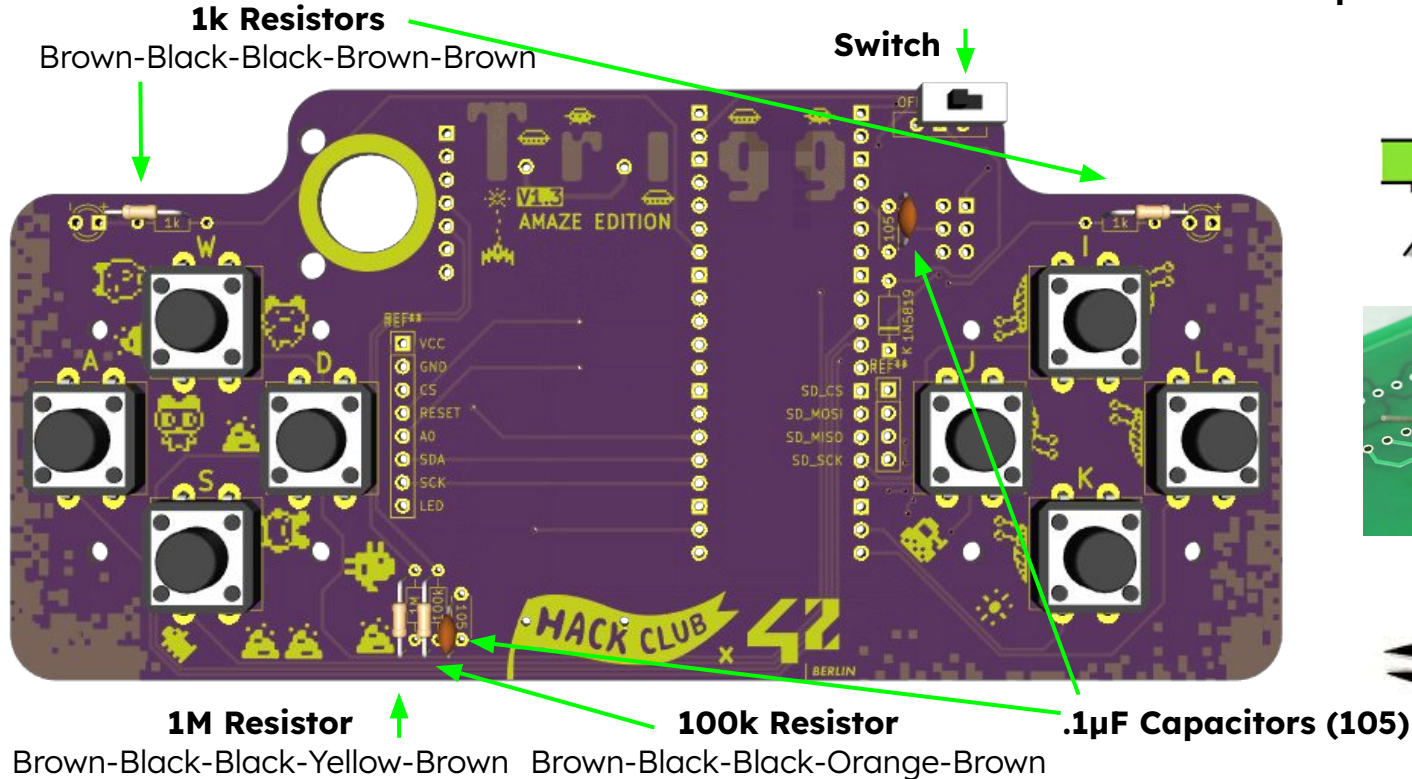
4: FRONT SIDE *BUTTONS*

Extra large pads for an easy start to soldering!



5: FRONT SIDE RESISTORS, CAPACITORS & SWITCH

Orientation of these components does not matter!

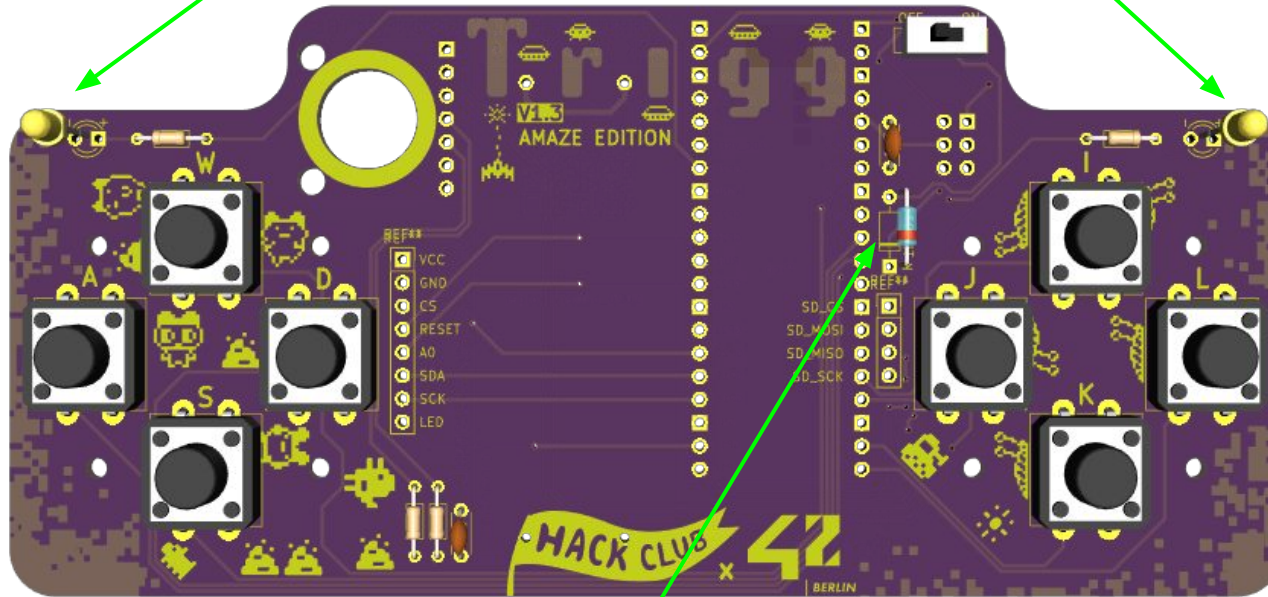


6: FRONT SIDE *LEDS & DIODE*

LEDs

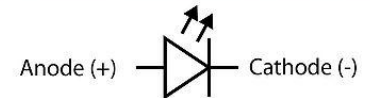
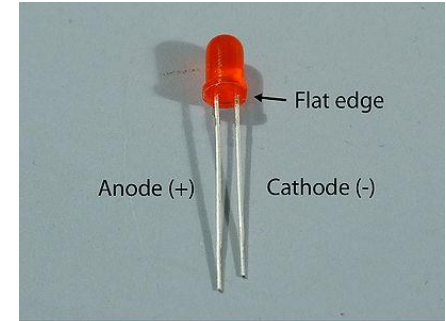
Short leg (cathode -) to left hole (circular pad)
Long leg (anode +) to right hole (square pad)

Orientation of these components matters!



1N5819 Diode

Stripe on diode should
match stripe on PCB

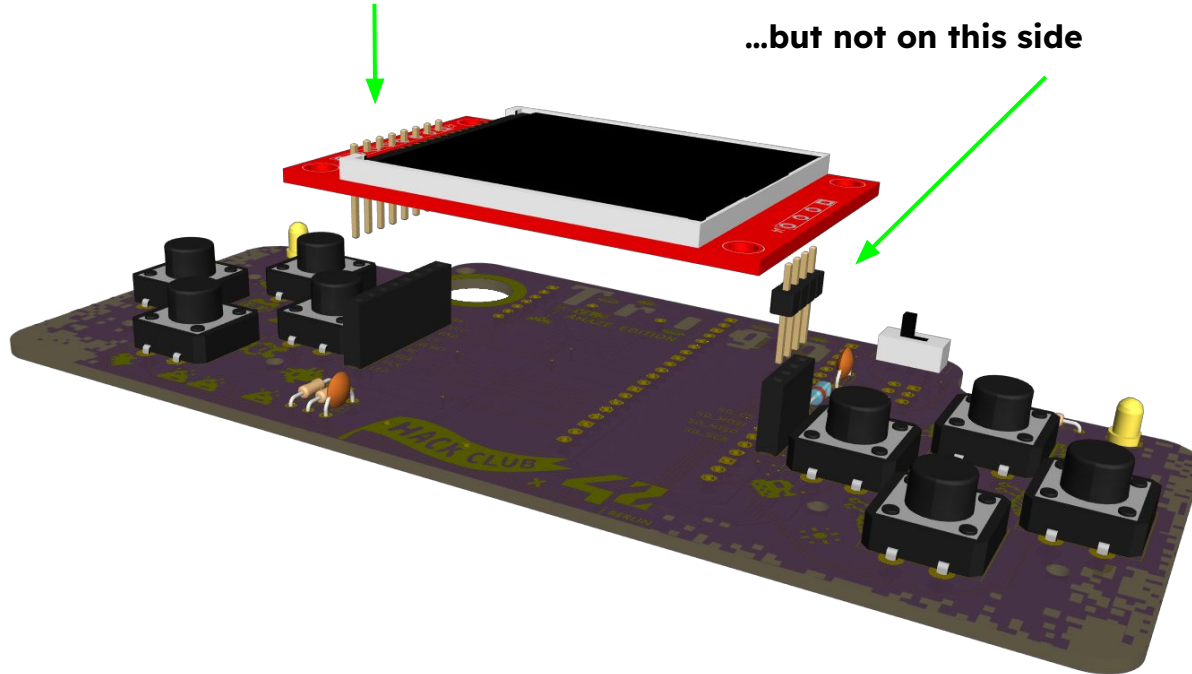


7: FRONT SIDE 1.8" TFT DISPLAY

Connect display to pin headers and
pin headers to sockets before soldering!

Pin header is pre-soldered on this side...

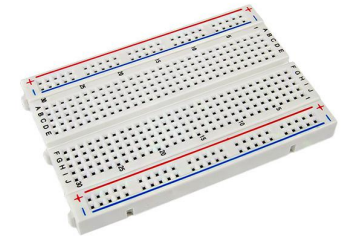
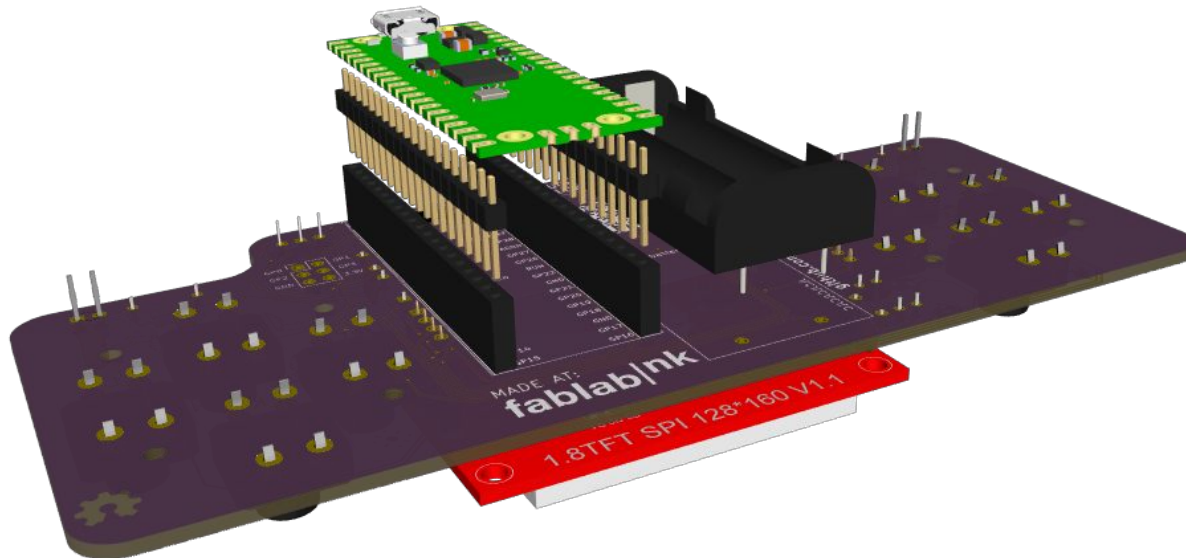
...but not on this side



8: BACK SIDE *PI PICO & BATTERY HOLDER*

Mount the pin headers on a breadboard to solder the Pico!

The Raspberry Pi Pico has many pins!
We can help to solder it if you're running out of time...

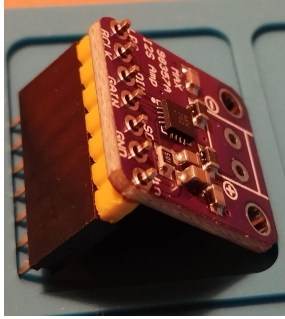


Don't forget to snip the
'dot' off from the back
of the battery holder
before soldering!

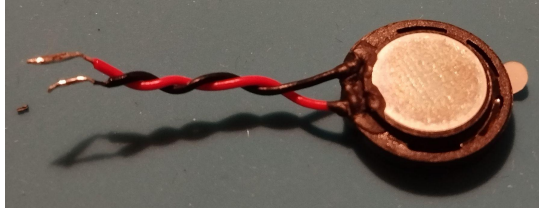


9: BACK SIDE *SPEAKER & AMPLIFIER BOARD*

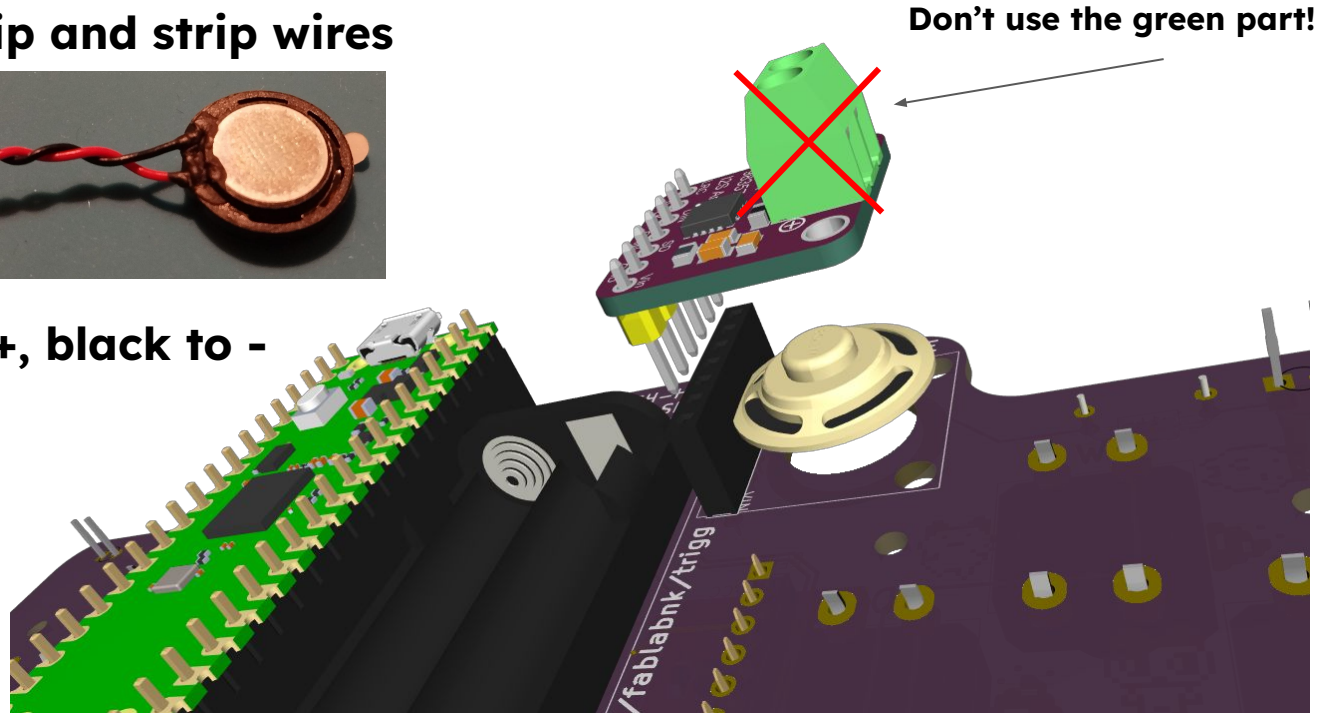
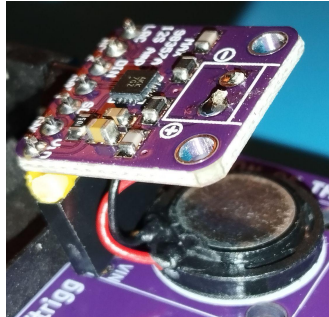
a. Solder pin header



b. Snip and strip wires



c. Solder red wire to +, black to -



10: FLASHING FIRMWARE

This part should have already been done for you!

If not, follow the instructions at:

<https://github.com/hackclub/sprig/blob/main/docs/UPLOAD.md>

Then connect Trigg to a USB charger or PC via micro USB cable

If all went well this message will appear on screen...



11: UPLOADING A GAME

This part should have already been done for you!
If not...

- Visit <https://sprig.hackclub.com/gallery>
- Choose a game and click on it
- Click drop-down arrow next to *Run* button and choose *Run on Device*

Check that screen, controls and sound are working correctly

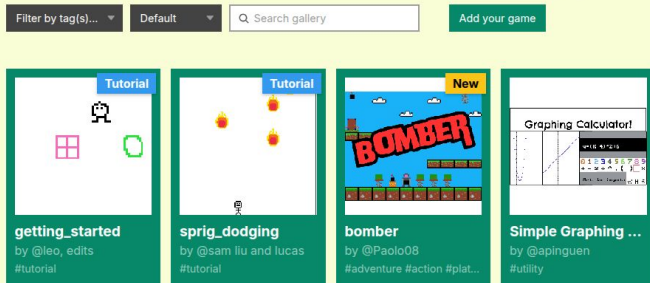
**Always test with USB cable
(partially charged batteries can
cause freezes)**

a.

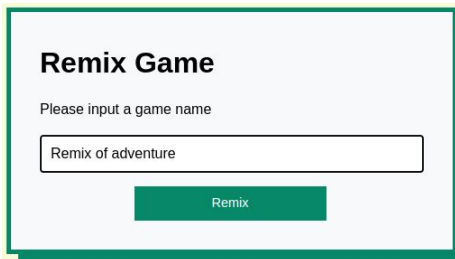
Gallery

We currently have 935 games made by Hack Clubbers. Click on any to play in the browser and hack on the code in our editor!

Add your game to the gallery and get your very own Sprig console.



b.



c.

